The learnability of sign language morphology: an artificial language learning study

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It has been suggested that the emergence of a linguistic system is facilitated by iconic form-meaning mappings, providing a solution to the symbol grounding problem (Cuskley & Kirby, 2018). Gauging the learnability of linguistic features when they are iconic vs. non-iconic can help us to understand the precise linguistic properties that may have been favored in the emergence and evolution of language. In sign language morphology, the handshape of classifier constructions has been found to be harder to learn than their orientation or movement (Marshall & Morgan, 2015). It is as yet unknown how the learnability of classifier constructions interact with iconicity. Drawing inspiration from artificial language learning experiments studying the effect of second language learning on morphological features (Atkinson et al., 2018; Smith, 2024), we present an artificial sign language learning experiment in which we systematically manipulate iconicity in both lexical signs and classifier constructions to better understand which features may have supported the learnability of morphological patterns in language emergence and evolution.

Adapted from Smith (2024), the stimuli in our experiment consist of scenes of six animals (bird, cow, crocodile, dog, fish, and giraffe) performing one of three movement types (moving horizontally, bouncing, or moving diagonally downward), resulting in 18 possible scenes. Participants were shown videos of possible British Sign Language (BSL) descriptions of these scenes: a lexical sign for the animal followed by a different classifier construction representing each of the different animals and their movement. 88 participants were recruited on Prolific and were randomly assigned to one of four conditions: BSL, counter-

iconic lexical, counter-iconic classifier, and counter-iconic both. Depending on the condition, the lexical sign and/or the classifier match or do not match the appropriate BSL signs (see Fig. 1). In the training phase, participants were shown 9 of the 18 scenes paired with their corresponding video description; participants first observed the scene and video, then saw the video and selected the corresponding scene from an array and finally they were given feedback (correct or incorrect) about their selection. In the testing phase, all 18 scenes were shown in a random order, and participants were asked to record a video description for each scene. This procedure was repeated four times.

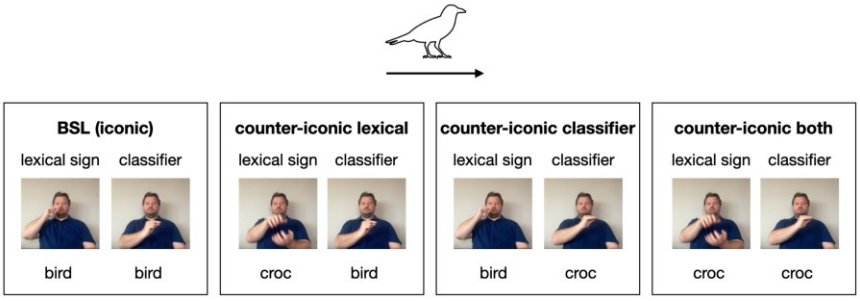

Figure 1. Examples of stimuli across the conditions for the stimulus *bird moving horizontally*. In the BSL condition, participants are trained ion the BSL lexical sign and classifier for bird; in the other conditions, either the lexical sign or classifier are swapped for the BSL sign for another referent (here, the crocodile); e.g. in the counter-iconic lexical condition participants are taught the lexical sign for crocodile paired with the classifier for bird.

We focus on the learnability of the classifier handshape. The proportion of correct classifier handshapes across all rounds is highest in the BSL condition (M=0.41; SD=0.27), followed by the counter-iconic both condition (M=0.32; SD=0.26) and the counter-iconic lexical condition (M=0.29; SD=0.25), with the lowest proportion of correct reproductions in the counter-iconic classifier condition (M=0.17; SD=0.19).

A generalized linear mixed-effects model with a binomial error distribution was run predicting the proportion of correct classifier handshape, with fixed effects for lexical iconicity, classifier construction iconicity, and their interaction. Random intercepts were included for participants and items. In addition, by-participant random slopes were specified for round. Lexical iconicity and classifier construction iconicity were deviation-coded. We find significant effects of an iconic classifier (β=0.75, *SE*=0.24, z=3.5, p<.01) and an iconic lexical sign by iconic classifier interaction (β=1.58, *SE*=0.49, z=3.23, p<.01), reflecting the very low accuracy in the counter-iconic classifier condition.

In sum, our findings support the known benefit of iconicity in second language learning (Ortega, 2017), and provide new insight into the interaction between iconicity and specific morphological features in sign languages. Interestingly, the counter-iconic both condition yielded higher accuracy rates than conditions with mixed iconic and counter-iconic components, indicating that iconicity does not operate in isolation, but interacts across linguistic levels.

References

Atkinson, M., Smith, K., & Kirby, S. (2018). Adult Learning and Language Simplification. *Cognitive Science*, *42*(8), 2818–2854. https://doi.org/10.1111/cogs.12686

Cuskley, C., & Kirby, S. (2018). Synesthesia, Cross-Modality, and Language Evolution. In J. Simner & E. Hubbard (Eds.), *Oxford Handbook of Synesthesia*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199603329.013.0043

Marshall, C. R., & Morgan, G. (2015). From Gesture to Sign Language: Conventionalization of Classifier Constructions by Adult Hearing Learners of British Sign Language. *Topics in Cognitive Science*, *7*(1), 61–80. https://doi.org/10.1111/tops.12118

Ortega, G. (2017). Iconicity and Sign Lexical Acquisition: A Review. *Frontiers in Psychology*, *8*, 1280. https://doi.org/10.3389/fpsyg.2017.01280

Smith, K. (2024). Simplifications made early in learning can reshape language complexity: An experimental test of the Linguistic Niche Hypothesis. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *46*(0). https://escholarship.org/uc/item/79862380